Wafer Lens

현장 문제

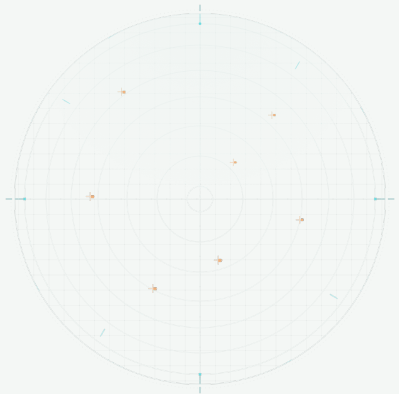
Wafer defect maps contain spatial patterns that may help organize follow-up inspection. This prototype explores whether a map can be matched to known visual pattern classes while making uncertainty and weak matches visible.

제안하는 도구

A browser workbench accepts sample maps or a CSV die map, lets the reviewer edit pass/fail cells, and shows the nearest labeled examples, relative support, uncertainty, and abstention. The public-data model uses a feature-based 9-nearest-neighbor vote; its pattern labels are not physical root-cause diagnoses.

LIVE PROTOTYPE

https://wafer-lens-lilac.vercel.app



CONCEPT ART · NOT A SCREENSHOT

WORKFLOW

01 샘플 또는 CSV map 불러오기

02 셀을 편집하며 가까운 패턴 비교

03 불확실성·기권을 확인하고 검토

패턴은 원인 진단이나 자동 판정이 아닙니다.

검증 결과 · V2

데이터와 방법

WM-811K wafer map · 9-nearest-neighbor classifier on 32x32 normalized die maps using failure-density, radial, angular, spatial, and adjacency features; inverse-square distance vote. Scores are relative neighbor support, not calibrated probabilities.

평가 방법

Deterministic, capped, class-stratified v2 evaluation by lotName. The v2 lot partition has no overlap with its own train/validation groups. Metrics describe this exploratory sample, not deployment prevalence or fab performance. Train 1340개 / 1001 lots, final test 351개 / 302 lots. 해시 기반 lot 분리: 학습 3·4, 검증 1, 최종 평가 2.

MEASURED RESULTS

정확도 · 기권 포함 212 / 351 final test wafers	**60.4%**
Macro F1 9개 class F1의 비가중 평균	**0.661**
기권 비율 80 / 351 wafers	**22.8%**
응답 정확도 212 / 271 non-abstained wafers	**78.2%**

낮은 recall class: Loc 20.0% · Scratch 25.0% · Edge-Loc 42.5%

수치는 입력된 평가 결과를 그대로 표시합니다. 운영 성능이나 수율 개선을 의미하지 않습니다.

재현성과 한계

재현

Node.js 24 or newer for model evaluation 환경에서 아래 명령을 실행합니다.

```
npm install
```

```
node scripts/data/evaluate.mjs
```

```
npm run build
```

데이터 출처

WM-811K wafer map · QINGYI · https://www.kaggle.com/datasets/qingyi/wm811k-wafer-map/metadata · CC0: Public Domain. Original upstream provenance and rights were not independently verified.

모델 선택

모델 선택: 검증 분할 358 wafers에서 학습 표본 700 / 1340개를 비교 (macro-F1 0.620 / 0.664). 최종 351 wafers (302 lots)는 v2 학습·검증 lots와 분리했지만, 최종 평가 ID 36개가 v1 학습에 포함되어 전체 개발 이력 기준의 완전 독립 시험은 아닙니다.

한계

Dataset labels are visual pattern classes, not physical root causes. · The holdout is small and class-stratified, with limited Near-full examples. · 32x32 nearest-neighbor downsampling may discard tiny defects. · A novel fab, process, image modality, or annotation policy can shift performance. · The source listing reports CC0 but original upstream rights have not been audited.

다음 검증

데이터 권리와 라벨 품질 확인, 다른 데이터 분할에서의 재현, 도메인 전문가 검토, 실제 제조 환경의 별도 검증이 필요합니다.